

Q. 1 (An interesting formula). Note that, for every positive integer x ,

$$\sum_{k=1}^{\infty} 1_{\{x \geq k\}} = \sum_{k=1}^x 1 = x.$$

Hence,

$$\sum_{k=1}^{\infty} \mathbf{P}\{X \geq k\} = \sum_{k=1}^{\infty} \mathbb{E}(1_{\{X \geq k\}}) = \mathbb{E}\left(\sum_{k=1}^{\infty} 1_{\{X \geq k\}}\right) = \mathbb{E}(X).$$

Q. 2 (A stupid game). a. Let N be the random variable corresponding to the number told to the person, D the random variable corresponding to the number rolled on the die and C the random variable corresponding to the side obtained when flipping the coin.

The die being fair, we have $\mathbb{P}(D = i) = \frac{1}{6}$ for all $i = 1, \dots, 6$. Similarly, as the coin is fair, we have $\mathbb{P}(C = H) = \mathbb{P}(C = T) = \frac{1}{2}$. From the question, we know that $(N|C = H) = D$ and $(N|C = T) = 2D$ and we are looking for $\mathbb{E}[N]$. By the law of total expectations,

$$\begin{aligned} \mathbb{E}[N] &= \mathbb{E}[N|C = H]\mathbb{P}(C = H) + \mathbb{E}[N|C = T]\mathbb{P}(C = T) \\ &= \frac{1}{2}\mathbb{E}[D] + \frac{1}{2}\mathbb{E}[2D] \\ &= \frac{3}{2}\mathbb{E}[D] \quad \text{by linearity of the expectation} \end{aligned}$$

We therefore need to compute $\mathbb{E}[D]$:

$$\begin{aligned} \mathbb{E}[D] &= \sum_{i=1}^6 i\mathbb{P}(D = i) \\ &= \frac{1 + 2 + 3 + 4 + 5 + 6}{6} = \frac{21}{6} = \frac{7}{2} \end{aligned}$$

By combining the results:

$$\mathbb{E}[N] = \frac{3}{2} \cdot \frac{7}{2} = \frac{21}{4}$$

Remark: An alternative way of doing this question and which is faster and more efficient is to set up the problem slightly differently: let N and D defined as above. We now denote by F the random variable that takes value 1 if the coin comes up heads and value 2 if the coin comes up tails: in other words, $F = 1$ with probability $\frac{1}{2}$ and $F = 2$ with probability $\frac{1}{2}$. As the coin flip and the die roll are independent, so are F and D . Note as well that $N = F \cdot D$.

$$\mathbb{E}[N] = \mathbb{E}[F \cdot D] = \mathbb{E}[F] \cdot \mathbb{E}[D] \quad \text{by independence}$$

As $\mathbb{E}[F] = \frac{1}{2} \cdot 1 + \frac{1}{2} \cdot 2 = \frac{3}{2}$ and from the same calculation as above $\mathbb{E}[D] = \frac{7}{2}$. We therefore obtain $\mathbb{E}[N] = \frac{3}{2} \cdot \frac{7}{2} = \frac{21}{4}$

b. Using the same notation as above, we compute using Bayes:

$$\begin{aligned}
 \mathbb{P}(C = H|N = 6) &= \frac{\mathbb{P}(N = 6|C = H)\mathbb{P}(C = H)}{\mathbb{P}(N = 6|C = H)\mathbb{P}(C = H) + \mathbb{P}(N = 6|C = T)\mathbb{P}(C = T)} \\
 &= \frac{\mathbb{P}(D = 6)\mathbb{P}(C = H)}{\mathbb{P}(D = 6)\mathbb{P}(C = H) + \mathbb{P}(D = 3)\mathbb{P}(C = T)} \\
 &= \frac{1/6 \cdot 1/2}{1/6 \cdot 1/2 + 1/6 \cdot 1/2} \\
 &= \frac{1}{2}
 \end{aligned}$$

In a similar way, we obtain:

$$\mathbb{P}(C = T|N = 6) = \frac{1}{2}$$

As a result, given only the information that we tell the person 6m the person cannot determine whether it is more likely that heads or tails came up (as both have equal probability).

Q. 3 (Bag of coins). a. Let Y denote the currency type and X denote the number of heads obtained in two independent flips of a coin. Note that,

$$\mathbb{P}\{Y = f\} = \frac{2}{3} \quad \text{and} \quad \mathbb{P}\{Y = c\} = \frac{1}{3}.$$

Since the coin is just flipped twice, there are three different outcomes for the number of heads. When the coin is fair, we have

$$\begin{aligned}
 \mathbb{P}\{X = 0|Y = f\} &= \frac{1}{4}, \\
 \mathbb{P}\{X = 1|Y = f\} &= \frac{1}{2}, \\
 \mathbb{P}\{X = 2|Y = f\} &= \frac{1}{4}.
 \end{aligned}$$

Then we can compute $\mathbb{E}[X|Y = f] = 1$ given a fair coin. When the coin is Canadian currency, similarly, we have

$$\begin{aligned}
 \mathbb{P}\{X = 0|Y = c\} &= 0.35^2, \\
 \mathbb{P}\{X = 1|Y = c\} &= 2 \cdot (0.65 \cdot 0.35) \\
 \mathbb{P}\{X = 2|Y = c\} &= 0.65^2
 \end{aligned}$$

Then we finally have $\mathbb{E}[X|Y = c] = 1.3$, when we have Canadian currency.

b. By tower property, we have

$$\mathbb{E}[X] = \mathbb{E}[X|Y = f]\mathbb{P}(Y = f) + \mathbb{E}[X|Y = c]\mathbb{P}(Y = c) = \frac{2}{3} + 1.3 \cdot \frac{1}{3} = \frac{11}{10}.$$

c. By Bayes,

$$\begin{aligned}
 \mathbb{P}(Y = c|X = 1) &= \frac{\mathbb{P}(X = 1|Y = c)\mathbb{P}(Y = c)}{\mathbb{P}(X = 1|Y = c)\mathbb{P}(Y = c) + \mathbb{P}(X = 1|Y = f)\mathbb{P}(Y = f)} \\
 &= \frac{2 \cdot 0.65 \cdot 0.35 \cdot \frac{1}{3}}{2 \cdot 0.65 \cdot 0.35 \cdot \frac{1}{3} + \frac{1}{2} \cdot \frac{2}{3}} = \frac{2 \cdot 0.65 \cdot 0.35 \cdot \frac{1}{3}}{2 \cdot 0.65 \cdot 0.35 + 1} \\
 &= \frac{0.455}{1.455} \simeq 0.31
 \end{aligned}$$

- Q. 4 (Love triangles).** a. If we fix a person, there are $n - 1$ potential hookups, each of them with a probability p of being a hookup. If we enumerate these $n - 1$ people as number 1, 2, ..., $n - 1$, we can write

$$\mathbb{E}[\text{number of hookups}] = \sum_{i=1}^{n-1} \mathbb{E}[1_{\{i \text{ is a hookup}\}}] = \sum_{i=1}^{n-1} \mathbb{P}(i \text{ is a hookup}) = p(n-1),$$

which gives us the expected number of hookups a given individual has at once.

- b. We calculate the probability that a given individual has no hookups or has exactly one hookup. If we denote these two events by E_0 and E_1 , we know

$$\mathbb{P}(E_0) = (1-p)^{n-1}, \quad \mathbb{P}(E_1) = (n-1)(1-p)^{n-2}p.$$

Let X denote the number of hookups an individual has at once. The desired probability of a given individual has more than one hookup is given by

$$\begin{aligned} \mathbb{P}\{X > 1\} &= 1 - \mathbb{P}\{X \leq 1\} \\ &= 1 - \mathbb{P}\{X = 0\} - \mathbb{P}\{X = 1\} \\ &= 1 - (1-p)^{n-1} - (n-1)(1-p)^{n-2}p. \end{aligned}$$

- c. In the following we will enumerate the n people on the site by 1 to n . If we use N to denote the random variable standing for the total number of love triangles on the site, we can write by linearity of expectation

$$\mathbb{E}[N] = \sum_{1 \leq i < j < k \leq n} \mathbb{E}[1_{\{i, j, k \text{ is a love triangle}\}}] = \sum_{1 \leq i < j < k \leq n} \mathbb{P}[i, j, k \text{ is a love triangle}].$$

However, if we fix three people, the probability that they form a love triangle is clearly given by p^3 . We thus conclude that the answer is given by

$$\binom{n}{3} p^3.$$

- Q. 5 (Overbooked flights).** a. The probability that the plane is overbooked is given by the probability that at least 201 people show up. Let E_1 denote the event that 201 people show up and E_2 the event that 202 people show up. Let us also denote $\{A_i\}_{i=1}^{202}$ the sequence of events such that A_i corresponds to the event that the i -th passenger shows up. By assumption, these A_i are mutually independent and we have:

$$\mathbb{P}(A_i) = \frac{98}{100}, \quad \mathbb{P}(A_i^C) = \frac{2}{100}$$

We can write E_1 as the collection of events such that only one passenger does not show up:

$$E_1 = \bigcup_{i=1}^{202} \left\{ \left(\bigcap_{k \neq i, 1 \leq k \leq 202} A_k \right) \cap A_i^C \right\}$$

and we note that the $\left\{ \left(\bigcap_{k \neq i} A_k \right) \cap A_i^C \right\}_{i=1}^{202}$ are mutually exclusive. Hence,

$$\mathbb{P}(E_1) = \sum_{i=1}^{202} \mathbb{P} \left(\left(\bigcap_{k \neq i} A_k \right) \cap A_i^C \right) \quad \text{by the third axiom of probability}$$

We can prove that the events $\{A_i\}_{i=1}^{202}$ stay mutually independent if you replace one of them by its complement. Without loss of generality, let us show that the events

$\{A_1^C, A_2, \dots, A_{202}\}$ are mutually independent.

Reminder Events $\{A_1, A_2, \dots, A_n\}$ are mutually independent iff for any sub-collection of $k \leq n$ events, $A_{i_1}, A_{i_2}, \dots, A_{i_k}$, it is true that:

$$\mathbb{P}(A_{i_1} \cap A_{i_2} \cap \dots \cap A_{i_k}) = \mathbb{P}(A_{i_1})\mathbb{P}(A_{i_2}) \dots \mathbb{P}(A_{i_k})$$

Proof. From the mutual independence, we know that this property is true for any sub-collection containing only sets of the form A_{i_k} , so we only need to check the property for sub-collections containing A_1^C . Let $k \leq n$.

$$\begin{aligned} \mathbb{P}(A_1^C \cap A_{i_2} \cap \dots \cap A_{i_k}) &= \mathbb{P}(A_{i_2} \cap \dots \cap A_{i_k}) - \mathbb{P}(A_1 \cap A_{i_2} \cap \dots \cap A_{i_k}) \\ &\quad \text{by third principle since } A_1 \cap A_1^C = \emptyset \\ &= \mathbb{P}(A_{i_2}) \dots \mathbb{P}(A_{i_k}) - \mathbb{P}(A_1)\mathbb{P}(A_{i_2}) \dots \mathbb{P}(A_{i_k}) \\ &\quad \text{by mutual independence} \\ &= (1 - \mathbb{P}(A_1))\mathbb{P}(A_{i_2}) \dots \mathbb{P}(A_{i_k}) \\ &= \mathbb{P}(A_1^C)\mathbb{P}(A_{i_2}) \dots \mathbb{P}(A_{i_k}) \end{aligned}$$

□

Let $i \in \{1, \dots, 202\}$ and let us compute:

$$\begin{aligned} \mathbb{P}\left(\left(\bigcap_{k \neq i} A_k\right) \cap A_i^C\right) &= \left(\prod_{k \neq i, 1 \leq k \leq 202} \mathbb{P}(A_k)\right) \times \mathbb{P}(A_i^C) \quad \text{by independence} \\ &= \left(\frac{98}{100}\right)^{201} \frac{2}{100} \end{aligned}$$

Hence:

$$\begin{aligned} \mathbb{P}(E_1) &= \sum_{i=1}^{202} \left(\frac{98}{100}\right)^{201} \frac{2}{100} \\ &= 202 \cdot \frac{2}{100} \cdot \left(\frac{98}{100}\right)^{201} \end{aligned}$$

where the last equality comes from the fact that we sum the same term 202 times.

Let us note that $E_2 = \bigcap_{i=1}^{202} A_i$ and by mutual independence, we have:

$$\mathbb{P}(E_2) = \mathbb{P}\left(\bigcap_{i=1}^{202} A_i\right) = \prod_{i=1}^{202} \mathbb{P}(A_i) = \left(\frac{98}{100}\right)^{202}$$

The probability that the plane is overbooked is hence given by

$$\mathbb{P}(E_1 \cup E_2) = 202 \frac{2}{100} \left(\frac{98}{100}\right)^{201} + \left(\frac{98}{100}\right)^{202}.$$

from the third principle since $E_1 \cap E_2 = \emptyset$

- b. Let X be the income in ticket fees for this flight and let us denote by N the number of passengers who show up. We want to compute $\mathbb{E}[X]$.

Note that :

- If $N \leq 200$, then $X = 500 \cdot N$
- If $N = 201$, then $X = 500 \cdot N - 500$
- If $N = 202$, then $X = 500 \cdot N - 1000$

Therefore:

$$X = 500N - 500 \cdot \mathbb{1}_{N=201} - 1000 \cdot \mathbb{1}_{N=202}$$

by linearity of the expectation:

$$\begin{aligned} \mathbb{E}(X) &= 500\mathbb{E}[N] - 500\mathbb{P}(N = 201) - 1000\mathbb{P}(N = 202) \\ &= 500\mathbb{E}[N] - 500\mathbb{P}(E_1) - 1000\mathbb{P}(E_2) \quad \text{by using the notations of Q5.a} \end{aligned}$$

Note that $N = \sum_{i=1}^{202} \mathbb{1}_{A_i}$, hence

$$\begin{aligned} \mathbb{E}[N] &= \sum_{i=1}^{202} \mathbb{E}[\mathbb{1}_{A_i}] \quad \text{by linearity of the expectation} \\ &= \sum_{i=1}^{202} \mathbb{P}(A_i) \\ &= 202 \cdot 0.98 \\ &= 197.96 \end{aligned}$$

In the end,

$$\begin{aligned} \mathbb{E}[X] &= 500 \cdot 202 \cdot 0.98 - 500 \cdot 202 \cdot (0.98)^{201} \cdot 0.02 - 1000 \cdot (0.98)^{202} \\ &\simeq 98,928.30 \end{aligned}$$

- c. By using the same notations as above and by denoting Y the total income made by the airline from operating this flight, we have:

$$Y = 500N - 800 \cdot \mathbb{1}_{N=201} - 1600 \cdot \mathbb{1}_{N=202} = X - 300 \cdot \mathbb{1}_{N=201} - 600 \cdot \mathbb{1}_{N=202}$$

Therefore, by linearity of the expectation,

$$\begin{aligned} \mathbb{E}[Y] &= \mathbb{E}[X] - 300\mathbb{P}(N = 201) - 600\mathbb{P}(N = 202) \\ &= \mathbb{E}[X] - 300\mathbb{P}(E_1) - 600\mathbb{P}(E_2) \quad \text{by using the notations of Q5.a and Q5.b} \\ &\simeq 98,928.30 - 300 \cdot 202 \cdot (0.98)^{201} \cdot 0.02 - 600 \cdot (0.98)^{202} \\ &\simeq 98,897.27 \end{aligned}$$